UNIVERSIDAD DEL VALLE DE GUATEMALA

CC3067 - REDES - 10

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PDF 2: Evidencias

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Tablas de enrutamiento

Salida de show ip route de los cinco routers, tomada después de configurar OSPF, EIGRP y la redistribución.

R-OSPF-1

```
R-OSPF-1>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 6 subnets
C       10.0.0.0 is directly connected, Serial1/0
C       10.0.0.4 is directly connected, Serial1/1
O       10.0.0.8 [110/20] via 10.0.0.6, 00:00:08, Serial1/1
O E2    10.0.0.12 [110/20] via 10.0.0.6, 00:00:08, Serial1/1
O E2    10.0.0.16 [110/20] via 10.0.0.6, 00:00:08, Serial1/1
O E2    10.0.0.20 [110/20] via 10.0.0.6, 00:00:08, Serial1/1
     172.16.0.0/16 is variably subnetted, 3 subnets, 3 masks
O E2    172.16.0.0/25 [110/20] via 10.0.0.6, 00:00:08, Serial1/1
O E2    172.16.0.128/26 [110/20] via 10.0.0.6, 00:00:08, Serial1/1
O E2    172.16.0.192/27 [110/20] via 10.0.0.6, 00:00:08, Serial1/1
O E2 192.168.0.0/20 [110/20] via 10.0.0.6, 00:00:08, Serial1/1
O    192.168.0.0/24 [110/21] via 10.0.0.2, 00:00:08, Serial1/0
                    [110/21] via 10.0.0.6, 00:00:08, Serial1/1
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/26 is directly connected, FastEthernet0/0.10
C       192.168.1.64/28 is directly connected, FastEthernet0/0.30

R-OSPF-1>
R-OSPF-1>
R-OSPF-1>
R-OSPF-1>
R-OSPF-1>
```

R-OSPF-2

```
R-OSPF-2>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 6 subnets
C       10.0.0.0 is directly connected, Serial1/0
O       10.0.0.4 [110/20] via 10.0.0.10, 00:01:13, Serial1/1
C       10.0.0.8 is directly connected, Serial1/1
O E2    10.0.0.12 [110/20] via 10.0.0.10, 00:01:13, Serial1/1
O E2    10.0.0.16 [110/20] via 10.0.0.10, 00:01:13, Serial1/1
O E2    10.0.0.20 [110/20] via 10.0.0.10, 00:01:13, Serial1/1
     172.16.0.0/16 is variably subnetted, 3 subnets, 3 masks
O E2    172.16.0.0/25 [110/20] via 10.0.0.10, 00:01:13, Serial1/1
O E2    172.16.0.128/26 [110/20] via 10.0.0.10, 00:01:13, Serial1/1
O E2    172.16.0.192/27 [110/20] via 10.0.0.10, 00:01:13, Serial1/1
O E2 192.168.0.0/20 [110/20] via 10.0.0.10, 00:01:03, Serial1/1
C    192.168.0.0/24 is directly connected, FastEthernet0/0
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
O       192.168.1.0/26 [110/21] via 10.0.0.1, 00:01:03, Serial1/0
                       [110/21] via 10.0.0.10, 00:01:03, Serial1/1
O       192.168.1.64/28 [110/21] via 10.0.0.1, 00:01:03, Serial1/0
                        [110/21] via 10.0.0.10, 00:01:03, Serial1/1

R-OSPF-2>
R-OSPF-2>
```

R-CENTRAL

```
R-CENTRAL>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 6 subnets
O       10.0.0.0 [110/30] via 10.0.0.9, 00:01:22, Serial1/1
                 [110/30] via 10.0.0.5, 00:01:22, Serial1/0
C       10.0.0.4 is directly connected, Serial1/0
C       10.0.0.8 is directly connected, Serial1/1
C       10.0.0.12 is directly connected, Serial1/2
C       10.0.0.16 is directly connected, Serial1/3
D       10.0.0.20 [90/640000] via 10.0.0.18, 00:01:44, Serial1/3
     172.16.0.0/16 is variably subnetted, 3 subnets, 3 masks
D       172.16.0.0/25 [90/386560] via 10.0.0.18, 00:01:46, Serial1/3
D       172.16.0.128/26 [90/642560] via 10.0.0.18, 00:01:44, Serial1/3
D       172.16.0.192/27 [90/642560] via 10.0.0.18, 00:01:44, Serial1/3
D    192.168.0.0/20 is a summary, 00:01:32, Null0
O    192.168.0.0/24 [110/11] via 10.0.0.9, 00:01:22, Serial1/1
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
O       192.168.1.0/26 [110/11] via 10.0.0.5, 00:01:32, Serial1/0
O       192.168.1.64/28 [110/11] via 10.0.0.5, 00:01:32, Serial1/0

R-CENTRAL>
```

R-EIGRP-1

```
R-EIGRP-1>
R-EIGRP-1>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 6 subnets
D EX    10.0.0.0 [170/665600] via 10.0.0.22, 00:01:42, Serial1/1
                 [170/1438720] via 10.0.0.13, 00:01:42, Serial1/0
D EX    10.0.0.4 [170/665600] via 10.0.0.22, 00:02:04, Serial1/1
                 [170/1438720] via 10.0.0.13, 00:02:03, Serial1/0
D EX    10.0.0.8 [170/665600] via 10.0.0.22, 00:02:04, Serial1/1
                 [170/1438720] via 10.0.0.13, 00:02:03, Serial1/0
C       10.0.0.12 is directly connected, Serial1/0
D       10.0.0.16 [90/640000] via 10.0.0.22, 00:02:04, Serial1/1
                  [90/1541120] via 10.0.0.13, 00:02:03, Serial1/0
C       10.0.0.20 is directly connected, Serial1/1
     172.16.0.0/16 is variably subnetted, 3 subnets, 3 masks
D       172.16.0.0/25 [90/514560] via 10.0.0.22, 00:02:04, Serial1/1
                      [90/1543680] via 10.0.0.13, 00:02:03, Serial1/0
C       172.16.0.128/26 is directly connected, FastEthernet0/0.10
C       172.16.0.192/27 is directly connected, FastEthernet0/0.20
D    192.168.0.0/20 [90/2553856] via 10.0.0.22, 00:01:52, Serial1/1
                    [90/3326976] via 10.0.0.13, 00:01:52, Serial1/0

R-EIGRP-1>
R-EIGRP-1>
```

R-EIGRP-2

```
R-EIGRP-2>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 6 subnets
D EX    10.0.0.0 [170/409600] via 10.0.0.17, 00:02:06, Serial1/0
D EX    10.0.0.4 [170/409600] via 10.0.0.17, 00:02:29, Serial1/0
D EX    10.0.0.8 [170/409600] via 10.0.0.17, 00:02:29, Serial1/0
D       10.0.0.12 [90/1541120] via 10.0.0.17, 00:02:29, Serial1/0
                  [90/1669120] via 10.0.0.21, 00:02:28, Serial1/1
C       10.0.0.16 is directly connected, Serial1/0
C       10.0.0.20 is directly connected, Serial1/1
     172.16.0.0/16 is variably subnetted, 3 subnets, 3 masks
C       172.16.0.0/25 is directly connected, FastEthernet0/0
D       172.16.0.128/26 [90/514560] via 10.0.0.21, 00:02:28, Serial1/1
D       172.16.0.192/27 [90/514560] via 10.0.0.21, 00:02:28, Serial1/1
D    192.168.0.0/20 [90/2297856] via 10.0.0.17, 00:02:16, Serial1/0

R-EIGRP-2>
R-EIGRP-2>
R-EIGRP-2>
R-EIGRP-2>
```

Evidencia del balanceo OSPF

```
R-OSPF-1>show ip ospf interface

FastEthernet0/0.10 is up, line protocol is up
  Internet address is 192.168.1.1/26, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 1.1.1.1, Interface address 192.168.1.1
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    No Hellos (Passive interface)
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
FastEthernet0/0.30 is up, line protocol is up
  Internet address is 192.168.1.65/28, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 1.1.1.1, Interface address 192.168.1.65
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    No Hellos (Passive interface)
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
Serial1/0 is up, line protocol is up
  Internet address is 10.0.0.1/30, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type POINT-TO-POINT, Cost: 20
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:06
  Index 3/3, flood queue length 0
```

```
  Index 3/3, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 2.2.2.2
  Suppress hello for 0 neighbor(s)
Serial1/1 is up, line protocol is up
  Internet address is 10.0.0.5/30, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type POINT-TO-POINT, Cost: 10
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:06
  Index 4/4, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 3.3.3.3
  Suppress hello for 0 neighbor(s)
R-OSPF-1>
R-OSPF-1>
R-OSPF-1>
R-OSPF-1>
```

```
R-OSPF-1>show ip route 192.168.0.0
Routing entry for 192.168.0.0/24
Known via "ospf 1", distance 110, metric 21, type intra area
  Last update from 10.0.0.2 on Serial1/0, 00:03:33 ago
  Routing Descriptor Blocks:
  * 10.0.0.2, from 2.2.2.2, 00:03:33 ago, via Serial1/0
      Route metric is 21, traffic share count is 1
    10.0.0.6, from 2.2.2.2, 00:03:33 ago, via Serial1/1
      Route metric is 21, traffic share count is 1

R-OSPF-1>
R-OSPF-1>
```

Evidencia del balanceo EIGRP

```
R-EIGRP-1>show ip eigrp topology all-links
IP-EIGRP Topology Table for AS 100/ID(172.16.0.193)

Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
       r - Reply status

P 10.0.0.0/30, 2 successors, FD is 665600
         via 10.0.0.22 (665600/409600), Serial1/1
         via 10.0.0.13 (1438720/281600), Serial1/0
P 10.0.0.4/30, 2 successors, FD is 665600
         via 10.0.0.22 (665600/409600), Serial1/1
         via 10.0.0.13 (1438720/281600), Serial1/0
P 10.0.0.8/30, 2 successors, FD is 665600
         via 10.0.0.22 (665600/409600), Serial1/1
         via 10.0.0.13 (1438720/281600), Serial1/0
P 10.0.0.12/30, 1 successors, FD is 1413120
         via Connected, Serial1/0
P 10.0.0.16/30, 2 successors, FD is 640000
         via 10.0.0.22 (640000/384000), Serial1/1
         via 10.0.0.13 (1541120/384000), Serial1/0
P 10.0.0.20/30, 1 successors, FD is 512000
         via Connected, Serial1/1
P 172.16.0.0/25, 2 successors, FD is 514560
         via 10.0.0.22 (514560/28160), Serial1/1
         via 10.0.0.13 (1543680/386560), Serial1/0
P 172.16.0.128/26, 1 successors, FD is 28160
         via Connected, FastEthernet0/0.10
P 172.16.0.192/27, 1 successors, FD is 28160
         via Connected, FastEthernet0/0.20
P 192.168.0.0/20, 2 successors, FD is 2553856
         via 10.0.0.22 (2553856/2297856), Serial1/1
         via 10.0.0.13 (3326976/2169856), Serial1/0
R-EIGRP-1>
R-EIGRP-1>
R-EIGRP-1>
R-EIGRP-1>
```

```
R-EIGRP-1>show ip protocols

Routing Protocol is "eigrp  100 "
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Default networks flagged in outgoing updates
  Default networks accepted from incoming updates
  EIGRP metric weight K1=1, K2=0, K3=1, K4=0, K5=0
  EIGRP maximum hopcount 100
  EIGRP maximum metric variance 4
Redistributing: eigrp 100
  Automatic network summarization is not in effect
  Maximum path: 4
  Routing for Networks:
     172.16.0.128/26
     172.16.0.192/27
     10.0.0.12/30
     10.0.0.20/30
  Passive Interface(s):
    FastEthernet0/0.10
    FastEthernet0/0.20
  Routing Information Sources:
    Gateway         Distance      Last Update
    10.0.0.13       90            8408
    10.0.0.22       90            8693
  Distance: internal 90 external 170

R-EIGRP-1>
R-EIGRP-1>
R-EIGRP-1>
R-EIGRP-1>
R-EIGRP-1>
```

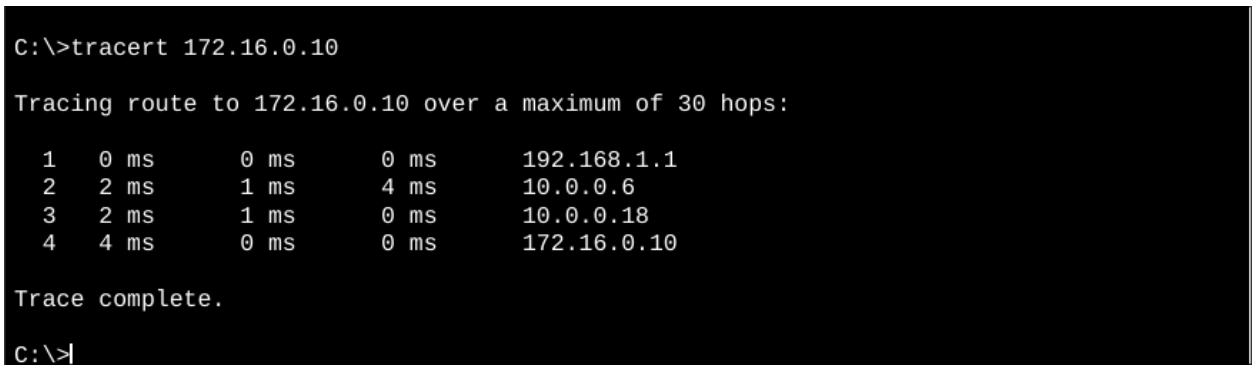
Prueba de conectividad extremo a extremo

```
C:\>ping 172.16.0.10

Pinging 172.16.0.10 with 32 bytes of data:

Request timed out.
Reply from 172.16.0.10: bytes=32 time=6ms TTL=125
Reply from 172.16.0.10: bytes=32 time=4ms TTL=125
Reply from 172.16.0.10: bytes=32 time=6ms TTL=125

Ping statistics for 172.16.0.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 6ms, Average = 5ms
```


C:\>tracert 172.16.0.10
Tracing route to 172.16.0.10 over a maximum of 30 hops:
1 0 ms 0 ms 0 ms 192.168.1.1
2 2 ms 1 ms 4 ms 10.0.0.6
3 2 ms 1 ms 0 ms 10.0.0.18
4 4 ms 0 ms 0 ms 172.16.0.10
Trace complete.
C:\>